

Green house DEA Analysis

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```
## Load packages
```

```
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
library(tidyr)
```

```
library(ggplot2)
```

```
library(Benchmarking) # standard radial DEA (Bogetoft & Otto)
```

```
## Warning: package 'Benchmarking' was built under R version 4.4.3
```

```
## Loading required package: lpSolveAPI
```

```
## Warning: package 'lpSolveAPI' was built under R version 4.4.3
```

```
## Loading required package: ucminf
```

```
## Loading required package: quadprog
```

```
library(deaR) # supports directional distance functions / undesirable outputs
```

```
## Warning: package 'deaR' was built under R version 4.4.3
```

```
## Please, to cite deaR in publications use:
```

```
## Bolos, V. J., Benitez, R., Coll-Serrano, V. (2025). deaR: Conventional and
```

```
## fuzzy DEA models with R. SoftwareX, 31, 102266, 1-6.
```

```
## DOI: 10.1016/j.softx.2025.102266
```

```
##
## Attaching package: 'deaR'
```

```
## The following object is masked from 'package:Benchmarking':
##
##      efficiencies
```

```
# set an output directory
output_dir <- "dea_outputs"
if (!dir.exists(output_dir)) dir.create(output_dir)
```

```
# dataset(simulated)
set.seed(42)
n_dmu <- 120  # was 40 -- raised so 8 variables can't trivially all reach the frontier

dmu_data <- tibble(
  dmu_id = paste0("Op_", sprintf("%03d", 1:n_dmu)),
  energy_kwh = round(rnorm(n_dmu, mean = 15000, sd = 4000)),
  water_l = round(rnorm(n_dmu, mean = 80000, sd = 20000)),
  labor_hours = round(rnorm(n_dmu, mean = 1200, sd = 300)),
  area_m2 = round(rnorm(n_dmu, mean = 2000, sd = 500)),
  capital_usd = round(rnorm(n_dmu, mean = 250000, sd = 60000))
)

# Each operation gets a latent "how well-run is this place" draw, which now
# genuinely drives both outputs (more) and spoilage (less) -- instead of
# yield/revenue/spoilage being independent of the inputs and of each other.
efficiency_factor <- rnorm(n_dmu, mean = 1.0, sd = 0.18)

dmu_data <- dmu_data %>%
  mutate(
    yield_kg = round((0.9*energy_kwh/15000 + 0.6*labor_hours/1200 + 0.5*area_m2/2000) *
      6000 * efficiency_factor + rnorm(n_dmu, 0, 1500)),
    revenue_usd = round(yield_kg * rnorm(n_dmu, mean = 5.0, sd = 0.6))
  )

crowding <- dmu_data$labor_hours / dmu_data$area_m2
spoilage_rate <- rbeta(n_dmu, 2, 20) * (1 + 0.3 * scale(crowding)[, 1]) / efficiency_factor
dmu_data$spoilage_rate <- round(pmax(pmin(spoilage_rate, 0.35), 0.01), 3)

dmu_data <- dmu_data %>% mutate(across(-c(dmu_id, spoilage_rate), ~ pmax(.x, 1)))

write.csv(dmu_data, file.path(output_dir, "simulated_greenhouse_data_v2.csv"), row.names = FALSE)
print(head(dmu_data))
```

```
## # A tibble: 6 × 9
##   dmu_id energy_kwh water_l labor_hours area_m2 capital_usd yield_kg revenue_usd
##   <chr>      <dbl>   <dbl>      <dbl>   <dbl>      <dbl>   <dbl>      <dbl>
## 1 Op_001    20484    50127        981    1576      236464    12607    68511
## 2 Op_002    12741    50591       1499    1456     134503    14324    65872
## 3 Op_003    16453    82494       1578    1758     163646    15123    65515
## 4 Op_004    17531    60067       1575    1832     161821    16250    77908
## 5 Op_005    16617    79964        786    1923     295712    10390    52260
## 6 Op_006    14576    71435       1815    1878     235383    19446    82906
## # i 1 more variable: spoilage_rate <dbl>
```

```
## Baseline DEA: desirable outputs only (ignores spoilage)
inputs_mat <- as.matrix(dmu_data[, c("energy_kwh", "water_l", "labor_hours",
                                     "area_m2", "capital_usd")])
outputs_mat <- as.matrix(dmu_data[, c("yield_kg", "revenue_usd")])

dea_vrs <- dea(X = inputs_mat, Y = outputs_mat,
               RTS = "vrs", ORIENTATION = "out")

dmu_data$efficiency_baseline <- eff(dea_vrs)

cat("\n--- Baseline VRS output-oriented efficiency (desirable outputs only) ---\n")
```

```
##
## --- Baseline VRS output-oriented efficiency (desirable outputs only) ---
```

```
print(summary(dmu_data$efficiency_baseline))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.000   1.000   1.164   1.288   1.430   2.836
```

```
dea_data_undesirable <- make_deadata(
  datadea = dmu_data,
  dmus    = "dmu_id",
  inputs  = c("energy_kwh", "water_l", "labor_hours", "area_m2", "capital_usd"),
  outputs = c("yield_kg", "revenue_usd", "spoilage_rate"),
  ud_outputs = 3
)
```

```
## Warning in make_deadata(datadea = dmu_data, dmus = "dmu_id", inputs = c("energy_kwh", : There are data with very different orders of magnitude. Try to redefine the units of measure or some linear problems may be ill-posed.
## (This is a warning, not an error.)
```

```
dea_undesirable <- model_basic(dea_data_undesirable, orientation = "oo", rts = "vrs")
eff_raw <- efficiencies(dea_undesirable)
dmu_data$efficiency_ddf <- 1 / eff_raw

cat("\n--- Waste-adjusted efficiency (undesirable-output model, 1 = frontier) ---\n")
```

```
##
## --- Waste-adjusted efficiency (undesirable-output model, 1 = frontier) ---
```

```
print(summary(dmu_data$efficiency_ddf))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.7507  0.9468  0.9816  0.9681  1.0000  1.0000
```

```
dmu_data <- dmu_data %>%
  mutate(
    efficient_baseline = efficiency_baseline <= 1.001, # <-- fixed, see note below
    efficient_ddf      = efficiency_ddf >= 0.999,
    rank_baseline      = rank(-efficiency_baseline),
    rank_ddf           = rank(-efficiency_ddf)
  ) %>%
  arrange(rank_ddf)
```

```
cat("\n Operations that look efficient WITHOUT waste penalty but aren't WITH it \n")
```

```
##
## Operations that look efficient WITHOUT waste penalty but aren't WITH it
```

```
print(dmu_data %>%
  filter(efficient_baseline == TRUE & efficient_ddf == FALSE) %>%
  select(dmu_id, efficiency_baseline, efficiency_ddf, spoilage_rate))
```

```
## # A tibble: 0 × 4
## # i 4 variables: dmu_id <chr>, efficiency_baseline <dbl>, efficiency_ddf <dbl>,
## #   spoilage_rate <dbl>
```

```
# Slack analysis
dea_slacks <- slack(X = inputs_mat, Y = outputs_mat, dea_vrs)

slack_summary <- data.frame(
  dmu_id      = dmu_data$dmu_id,
  energy_slack = dea_slacks$sx[, 1],
  water_slack  = dea_slacks$sx[, 2],
  labor_slack  = dea_slacks$sx[, 3]
) %>% arrange(desc(energy_slack))

cat("\n--- Operations with the largest energy slack (excess energy use) ---\n")
```

```
##
## --- Operations with the largest energy slack (excess energy use) ---
```

```
print(head(slack_summary, 10))
```

##	dmu_id	energy_slack	water_slack	labor_slack
## 1	Op_014	10386.961	0.0000	290.9111
## 2	Op_001	8657.133	0.0000	0.0000
## 3	Op_032	8117.204	9915.1334	0.0000
## 4	Op_059	7135.686	554.0305	0.0000
## 5	Op_079	6200.009	0.0000	100.7242
## 6	Op_012	5523.946	8724.2421	0.0000
## 7	Op_105	4872.701	0.0000	114.8347
## 8	Op_019	4371.975	6843.9527	0.0000
## 9	Op_087	3983.049	7431.4355	0.0000
## 10	Op_025	3863.956	17308.3276	130.4809

```
# Visualization
p1 <- ggplot(dmu_data, aes(x = efficiency_baseline, y = efficiency_ddf,
                           color = spoilage_rate, size = area_m2)) +
  geom_point(alpha = 0.75) +
  scale_color_viridis_c(name = "Spoilage\trate") +
  labs(
    title = "Baseline vs. Waste-Adjusted Efficiency",
    subtitle = "Points below the diagonal look efficient only when spoilage is ignored",
    x = "Efficiency score (desirable outputs only)",
    y = "Efficiency score (waste-adjusted)",
    size = "Area (m²)"
  ) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray50") +
  theme_minimal(base_size = 13)

ggsave(file.path(output_dir, "efficiency_comparison_plot.png"), p1,
        width = 8, height = 6, dpi = 200)
```

```
p2 <- ggplot(dmu_data, aes(x = reorder(dmu_id, -efficiency_ddf), y = efficiency_ddf)) +
  geom_col(aes(fill = spoilage_rate)) +
  coord_flip() +
  scale_fill_viridis_c(name = "Spoilage\trate") +
  labs(
    title = "Waste-Adjusted Efficiency Ranking by Operation",
    x = NULL, y = "Efficiency score (1 = frontier, lower = less efficient)"
  ) +
  theme_minimal(base_size = 10) +
  theme(axis.text.y = element_text(size = 6))

ggsave(file.path(output_dir, "efficiency_ranking_plot.png"), p2,
        width = 8, height = 10, dpi = 200)
```

```
# Export outputs
write.csv(dmu_data, file.path(output_dir, "dea_results_full.csv"), row.names = FALSE)
write.csv(slack_summary, file.path(output_dir, "dea_slack_analysis.csv"), row.names = FALSE)

cat("\nDone. Outputs written to:", normalizePath(output_dir), "\n",
    "- simulated_greenhouse_data.csv (input dataset)\n",
    "- dea_results_full.csv (efficiency scores, both models)\n",
    "- dea_slack_analysis.csv (input excess by operation)\n",
    "- efficiency_comparison_plot.png\n",
    "- efficiency_ranking_plot.png\n")
```

```
##
## Done. Outputs written to: C:\Users\91934\OneDrive\Documents\dea_outputs
## - simulated_greenhouse_data.csv (input dataset)
## - dea_results_full.csv (efficiency scores, both models)
## - dea_slack_analysis.csv (input excess by operation)
## - efficiency_comparison_plot.png
## - efficiency_ranking_plot.png
```